WORKED SOLUTION

STEP I 2006, Question 6

Pure • Pell Equation Transformations

STEP I 2006, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q6

The key idea

Each part asks whether a linear change of coordinates preserves a quadratic expression. The direct route is to substitute, expand and compare coefficients of a^2, ab and b^2. Behind the scenes there is a neater picture: $x^2 - 2y^2$ is the "size" of the number $x + y\sqrt{2}$, and multiplying by $3 + 2\sqrt{2}$ (whose size is $9 - 8 = 1$) leaves that size unchanged. Seeing this tells you exactly how to build the map in (iii).

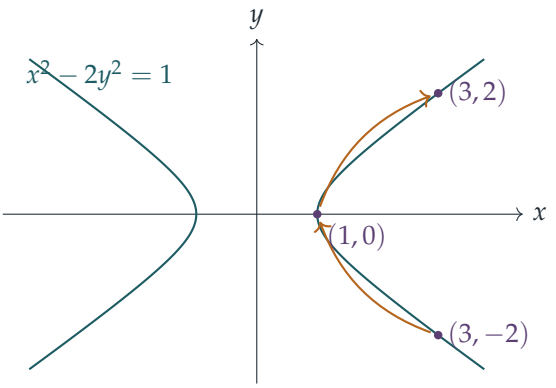
Solution

Part (i)

Suppose $a^2 - 2b^2 = 1$. Then

$$\begin{aligned}(3a + 4b)^2 - 2(2a + 3b)^2 &= (9a^2 + 24ab + 16b^2) - 2(4a^2 + 12ab + 9b^2) \\ &= a^2 - 2b^2 \\ &= 1,\end{aligned}$$

so $(3a + 4b,\ 2a + 3b)$ lies on $x^2 - 2y^2 = 1$, as required. In fact the first two lines are an identity, true for every a and b: the map preserves $x^2 - 2y^2$ everywhere, not just on the curve.



The map slides points along the right-hand branch: $(3,-2) \mapsto (1,0) \mapsto (3,2) \mapsto (17,12) \mapsto \cdots$.

Part (ii)

Suppose $Ma^2 - Nb^2 = 1$. Expanding,

$$M(5a+6b)^2 - N(4a+5b)^2 = (25M-16N)a^2 + (60M-40N)ab + (36M-25N)b^2.$$

The image point lies on the curve exactly when this equals $1 = Ma^2 - Nb^2$. Subtracting $Ma^2 - Nb^2$, the condition becomes

$$\begin{aligned} 0 &= (24M-16N)a^2 + (60M-40N)ab + (36M-24N)b^2 \\ &= 4(3M-2N)(2a^2+5ab+3b^2) \\ &= 4(3M-2N)(2a+3b)(a+b). \end{aligned}$$

Ask yourself Do we really need $3M = 2N$, or could the other factors be zero instead?

The condition has to hold for *every* point on the curve. When $M, N > 0$ the curve $Mx^2 - Ny^2 = 1$ is a hyperbola with infinitely many points. The lines $2x + 3y = 0$ and $x + y = 0$ each meet it in at most two points, so most points on the curve have $(2a+3b)(a+b) \neq 0$. Hence we need $3M - 2N = 0$. Conversely, if $3M = 2N$ the condition holds at every point.

So $3M = 2N$. Then $2 \mid 3M$, so M is even. The smallest choice is $M = 2$, which gives $N = 3$. As a final check,

$$2(5a+6b)^2 - 3(4a+5b)^2 = (50-48)a^2 + (120-120)ab + (72-75)b^2 = 2a^2 - 3b^2.$$

The smallest positive integers are $M = 2$ and $N = 3$.

Part (iii)

We want positive integers with

$$(Pa+Qb)^2 - 3(Ra+Sb)^2 = a^2 - 3b^2 \quad \text{for all } a, b,$$

which certainly makes the image of every point of the curve lie on it. Comparing coefficients:

$$P^2 - 3R^2 = 1, \qquad PQ = 3RS, \qquad Q^2 - 3S^2 = -3.$$

The first equation asks for a positive integer point (P, R) on $x^2 - 3y^2 = 1$. The smallest is $(2, 1)$, so try $P = 2$, $R = 1$. The second equation then gives $Q = \frac{3}{2}S$, and the third becomes $\frac{9}{4}S^2 - 3S^2 = -3$, that is, $S^2 = 4$. So $S = 2$ and $Q = 3$:

$$P = 2, \quad Q = 3, \quad R = 1, \quad S = 2.$$

Check: $(2a + 3b)^2 - 3(a + 2b)^2 = 4a^2 + 12ab + 9b^2 - 3a^2 - 12ab - 12b^2 = a^2 - 3b^2 = 1$. So $(2a + 3b,\ a + 2b)$ lies on $x^2 - 3y^2 = 1$.

Ask yourself Where does a map like this come from?

Write $(a + b\sqrt{3})(2 + \sqrt{3}) = (2a + 3b) + (a + 2b)\sqrt{3}$. The norm $x^2 - 3y^2$ of $x + y\sqrt{3}$ is multiplicative, and $2 + \sqrt{3}$ has norm $4 - 3 = 1$, so multiplying by it preserves the norm. Part (i) works the same way with $3 + 2\sqrt{2}$. For (ii), $(a\sqrt{2} + b\sqrt{3})(5 + 2\sqrt{6}) = (5a + 6b)\sqrt{2} + (4a + 5b)\sqrt{3}$, which is why the form $2x^2 - 3y^2$ appears there. Other answers are valid in (iii): multiplying by $(2 + \sqrt{3})^2 = 7 + 4\sqrt{3}$ gives $(7a + 12b,\ 4a + 7b)$.

Checks

- Start from $(1, 0)$ on $x^2 - 3y^2 = 1$. Repeating the map in (iii) gives $(2, 1)$, $(7, 4)$, $(26, 15)$, $(97, 56)$, and each satisfies $x^2 - 3y^2 = 1$ (for example $26^2 - 3 \cdot 15^2 = 676 - 675 = 1$).
- For (ii), $(1/\sqrt{2}, 0)$ is on $2x^2 - 3y^2 = 1$ and maps to $(5/\sqrt{2},\ 4/\sqrt{2})$. Then $2 \cdot \frac{25}{2} - 3 \cdot \frac{16}{2} = 25 - 24 = 1$.
- A brute-force search over $1 \le M, N < 30$, testing actual points of each hyperbola, confirms that $(2, 3)$ is the smallest pair that works.

Common slips

- In (ii), setting only the ab coefficient to zero without checking that the a^2 and b^2 coefficients then come out as M and $-N$. They do here, but that has to be shown.
- Saying "the identity must hold for all a, b" without justification, when the question only gives points on the curve. You need an argument, such as the factorisation above, that rules out the other factors vanishing.

Beyond STEP

These are Pell equations $x^2 - dy^2 = 1$. Their integer solutions are generated by powers of a fundamental unit $x_1 + y_1\sqrt{d}$ in the ring $\mathbb{Z}[\sqrt{d}]$, and the maps here are multiplication

by such units. In matrix language, each map is a 2×2 integer matrix that preserves a quadratic form, an element of an "integral orthogonal group". Davenport's *The Higher Arithmetic* has a lovely chapter-length treatment of Pell's equation via continued fractions.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.